

Missouri S&T

MISSOURI UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Section 4.5

The Dimension of a Vector Space

A Theorem from Section 4.4

Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a basis for a vector space V . This vector space V might not be $\mathbb{R}^n$. Then, the coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_B$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

Recall that a one-to-one linear transformation from a vector space V onto a vector space W is called an isomorphism from V onto W .

Theorems

If a vector space V has basis $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.

If a vector space V has a basis with n vectors, then every basis of V must have exactly n vectors.

The Dimension of a Vector Space

If V is spanned by a finite set, then V is said to be finite-dimensional and the dimension of V ($\dim V$) is the number of vectors in a basis for V .

The dimension of the zero vector space $\{0\}$ is defined to be zero.

If V is not spanned by a finite set, then V is said to be infinite-dimensional.

Example 1

Find a basis and state the dimension of the vector space.

$$V = \left\{ \begin{bmatrix} a - 4b - 2c \\ 2a + 5b - 4c \\ -a + 2c \\ -3a + 7b + 6c \end{bmatrix} : a, b, c \text{ real} \right\}$$

The Dimension of a Subspace

Let H be a subspace of a finite-dimensional vector space V . Any linearly independent set in H can be expanded (if necessary) to form a basis for H . Additionally, H is finite-dimensional and $\dim H \leq \dim V$

Example 2

The first four Laguerre polynomials are 1 , $1 - t$, $2 - 4t + t^2$, and $6 - 18t + 9t^2 - t^3$. Show that these polynomials form a basis B for $\mathbb{P}_3$.

Then, find the coordinate vector for

$$\mathbf{p}(t) = 7 - 8t + 3t^2$$

relative to the basis B .

Example 3

True or False.

1. If $\dim V = n$ and S is a linearly independent set in V , then S is a basis for V .
2. If there exists a linearly dependent set $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ in V , then $\dim V \leq p$.

The Rank of a Matrix

The rank of an $m \times n$ matrix A , denoted $\text{rank } A$, is the dimension of the column space of A .

The dimension of $\text{Col } A$ is the number of pivot columns of A .

Since the dimension of $\text{Row } A$ is the number of pivot rows of A , it is also equal to $\text{rank } A$.

The Nullity of a Matrix

The nullity of an $m \times n$ matrix A , denoted $\text{nullity } A$, is the dimension of the null space of A .

The dimension of $\text{Nul } A$ is the number of free variables in the equation $A\mathbf{x} = \mathbf{0}$.

Example 4

Determine the dimensions of $\text{Nul } A$, $\text{Col } A$, and $\text{Row } A$.

$$A = \begin{bmatrix} 1 & 0 & 9 & 1 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The Rank Theorem

The dimensions of the column space and the row space of an $m \times n$ matrix A are equal. This common dimension (the rank of A) equals the number of pivot positions in A and satisfies the equation

$$\text{rank } A + \text{nullity } A = n$$

Example 5

If the null space of a 5×6 matrix A is four-dimensional, what is the dimension of the row space of A ?

Example 6

- a) If A is a 6×8 matrix, what is the smallest possible dimension of $\text{Nul } A$?
- b) If A is a 6×4 matrix, what is the smallest possible dimension of $\text{Nul } A$?

Example 7

Suppose a nonhomogeneous system of six linear equations in eight unknowns has a solution, with two free variables. Is it possible to change some constants on the equations' right sides to make the new system inconsistent? Explain.

Example 8

Is it possible that all solutions of a homogeneous system of ten linear equations in twelve variables are multiples of one fixed nonzero solution?

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent.

- A is an invertible matrix.
- A is row equivalent to I_n .
- A has n pivot positions.
- The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- The columns of A form a linearly independent set.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent.

- The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
- The columns of A span $\mathbb{R}^n$.
- The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
- There is an $n \times n$ matrix C such that $CA = I$.
- There is an $n \times n$ matrix D such that $AD = I$.
- A^T is an invertible matrix.

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix. Then the following statements are equivalent.

- m. The columns of A form a basis for $\mathbb{R}^n$.
- n. $\text{Col } A = \mathbb{R}^n$.
- o. $\text{rank } A = n$.
- p. $\text{nullity } A = 0$
- q. $\text{Nul } A = \{\mathbf{0}\}$.

We could add one more statement that the text doesn't:
 $\det A \neq 0$
